National
Qualifications
2019

2019 Mathematics

Higher Paper 1 (Non-calculator)

Finalised Marking Instructions

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General marking principles for Higher Mathematics

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

For each question, the marking instructions are generally in two sections:

- *generic scheme — this indicates why each mark is awarded*
- *illustrative scheme — this covers methods which are commonly seen throughout the marking*

In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.

(c) One mark is available for each •. There are no half marks.

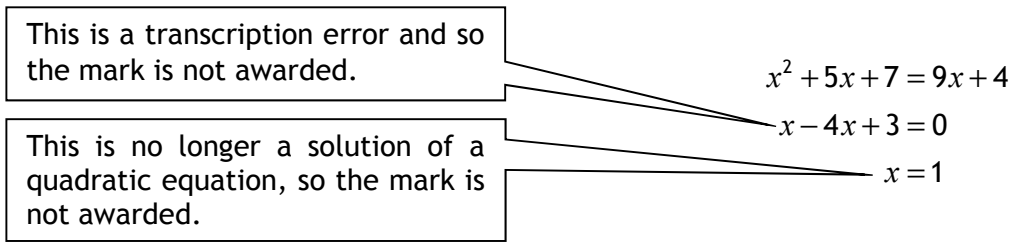
(d) If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.

(e) Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.

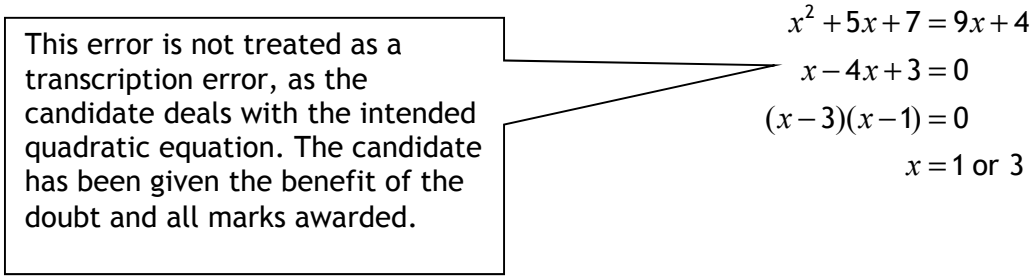
(f) Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.

(g) If an error is trivial, casual or insignificant, for example $6 \times 6 = 12$, candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.

(h) If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example



The following example is an exception to the above



(i) **Horizontal/vertical marking**
If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

	$\bullet^5$	$\bullet^6$
$\bullet^5$	$x=2$	$x=-4$
$\bullet^6$	$y=5$	$y=-7$

Horizontal: $\bullet^5$ $x=2$ and $x=-4$
$\bullet^6$ $y=5$ and $y=-7$

Vertical: $\bullet^5$ $x=2$ and $y=5$
$\bullet^6$ $x=-4$ and $y=-7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$\frac{15}{12}$ must be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$

$\frac{43}{1}$ must be simplified to 43

$\frac{15}{0{\cdot}3}$ must be simplified to 50

$\frac{4/5}{3}$ must be simplified to $\frac{4}{15}$

$\sqrt{64}$ must be simplified to 8*

*The square root of perfect squares up to and including 100 must be known.

(k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

(l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example

  $\left(x^3+2x^2+3x+2\right)(2x+1)$ written as

  $\left(x^3+2x^2+3x+2\right)\times 2x+1$

  $=2x^4+5x^3+8x^2+7x+2$

  gains full credit

- repeated error within a question, but not between questions or papers

(m) In any 'Show that…' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

(n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.

(o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.

(p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

Strategy 1 attempt 1 is worth 3 marks.	Strategy 2 attempt 1 is worth 1 mark.
Strategy 1 attempt 2 is worth 4 marks.	Strategy 2 attempt 2 is worth 5 marks.
From the attempts using strategy 1, the resultant mark would be 3.	From the attempts using strategy 2, the resultant mark would be 1.

In this case, award 3 marks.

Marking instructions for each question

Question	Generic scheme	Illustrative scheme	Max mark
1.	•1 start to differentiate	•1 $2x^3 \ldots$ or $\ldots -6x^2$	4
	•2 complete derivative and equate to 0	•2 $2x^3 - 6x^2 = 0$	
	•3 factorise derivative	•3 $2x^2(x-3)$	
	•4 process cubic for x	•4 0 and 3	

Notes:

1. •2 is only available if '$=0$' appears at either •2 or •3 stage, however see Candidate A.
2. Accept $2x^3 = 6x^2$ for •2.
3. Accept $x^2(2x-6)$ for •3.
4. For candidates who divide by x or x^2 throughout see Candidate B.
5. •3 is available to candidates who factorise **their** derivative from •2 as long as it is of equivalent difficulty.
6. $x=0$ and $x=3$ must be supported by valid working for •4 to be awarded.

Commonly Observed Responses:

Candidate A

Stationary points when $\frac{dy}{dx}=0$

$\frac{dy}{dx}=2x^3-6x^2$ •1 ✓ •2 ✓

$\frac{dy}{dx}=2x^2(x-3)$ •3 ✓

$x=0$ and $x=3$ •4 ✓

Candidate B

$2x^3-6x^2=0$ •1 ✓ •2 ✓

$2x^3=6x^2$ •3 ^

$x=3$ •4 ✗

Dividing by x^2 is not valid as $x=0$ is a solution.

Question	Generic scheme	Illustrative scheme	Max mark
2.	•1 use discriminant	•1 $(k-5)^2-4\times1\times1$	**3**
	•2 apply condition and simplify	•2 $k^2-10k+21=0$ or $(k-5)^2=4$	
	•3 determine values of k	•3 3, 7	

Notes:

1. Accept $(k-5)^2-4$ for •1.
2. Where candidates state an incorrect condition •2 is not available.
   •3 is available for finding the roots of the quadratic. See Candidate B.
3. Where x appears in any expression, no further marks are available.

Commonly Observed Responses:

Candidate A

For equal roots $b^2-4ac=0$

$(k-5)^2-4\times1\times1$ •1 ✓

$k^2-10k+21$ •2 ✓

$k=3,\ 7$ •3 ✓

Candidate B

For equal roots $b^2-4ac>0$ •2 ✗

$(k-5)^2-4\times1\times1$ •1 ✓

$k^2-10k+21=0$ or $(k-5)^2=4$

$k=3,\ 7$ •3 ✓ 1

Candidate C

$(k-5)^2-4\times1\times1=0$ •1 ✓

$k^2-10k=-21$ •2 ✓

$k=3,\ 7$ •3 ✓

No requirement for standard quadratic form

Question			Generic scheme	Illustrative scheme	Max mark
3.			•1 find radius of circle C_1 •2 state equation of circle C_2	•1 6 stated or implied by •2 •2 $(x-4)^2+(y+2)^2=36$	**2**

Notes:

1. Accept $\sqrt{3^2+1^2+26}=6$ or $\sqrt{-3^2+-1^2+26}=6$ for •1.
2. Do not accept $\sqrt{-3^2-1^2+26}=6$ for •1.
3. Do not accept $(x-4)^2+(y+2)^2=6^2$ for •2.
4. For candidates whose working for g^2+f^2-c does not arrive at a positive value, no marks are available. See Candidate A

Commonly Observed Responses:

Candidate A – 'fudging' negative values

$\sqrt{3^2+1^2-26}=4$ •1 ✗ •2 ✗

$(x-4)^2+(y+2)^2=16$

Question			Generic scheme	Illustrative scheme	Max mark
4.	(a)		•1 interpret recurrence relation •2 interpret recurrence relation •3 find m and c	•1 $9=6m+c$ •2 $11=9m+c$ •3 $m=\frac{2}{3}$ and $c=5$	**3**
Notes:					
1. Correct answer with no working award 0/3. 2. Do not penalise $9=m6+c$ or $11=m9+c$ at •1 and •2. 3. For candidates who state $m=\frac{2}{3}$, $c=5$ and then verify that these values work for the given terms, award 2/3.					
Commonly Observed Responses:					
	(b)		•4 calculate term	•4 $\frac{37}{3}$ or $12\frac{1}{3}$	**1**
Notes:					
4. The answer in (b) must be consistent with the values found in (a). 5. Accept $12\cdot\dot{3}$ or $12\cdot3...$ for •4. Do not accept a rounded answer.					
Commonly Observed Responses:					

Question			Generic scheme	Illustrative scheme	Max mark
5.	(a)		•1 find an appropriate vector eg $\overrightarrow{AB}$	•1 eg $\overrightarrow{AB}=\begin{pmatrix}3\\-6\\3\end{pmatrix}$	**3**
			•2 find a second vector eg $\overrightarrow{BC}$ and compare	•2 eg $\overrightarrow{BC}=\begin{pmatrix}4\\-8\\4\end{pmatrix}$ $\therefore \overrightarrow{AB}=\frac{3}{4}\overrightarrow{BC}$	
			•3 appropriate conclusion	•3 ... $\Rightarrow$ AB is parallel to BC (common direction) and B is a common point $\Rightarrow$ A,B and C are collinear.	

Notes:

1. Do not penalise inconsistent vector notation (eg lack of arrows or brackets).
2. Where •2 is not awarded, if a candidate states that $\overrightarrow{AB}=\overrightarrow{BC}$, only •1 is available.
3. •3 can only be awarded if a candidate has stated 'parallel', 'common point' and 'collinear'.
4. Candidates who state that 'points are parallel' or 'vectors are collinear' or 'parallel and share common point $\Rightarrow$ collinear' do not gain •3. There must be reference to points A, B and C.
5. Do not accept 'a, b and c are collinear' at •3.

Commonly Observed Responses:

Candidate A – missing labels

$\begin{pmatrix}3\\-6\\3\end{pmatrix}$ •1 ^

$\begin{pmatrix}4\\-8\\4\end{pmatrix}$ $\therefore \overrightarrow{AB}=\frac{3}{4}\overrightarrow{BC}$ •2 ✓ 1

Missing labels at •2 is a repeated error

$\Rightarrow$ AB is parallel to BC and B is a common point

$\Rightarrow$ A, B and C are collinear •3 ✓ 1

Candidate B

$\overrightarrow{AB}=\begin{pmatrix}3\\-6\\3\end{pmatrix}$ •1 ✓

$\overrightarrow{BC}=\begin{pmatrix}4\\-8\\4\end{pmatrix}$

$\begin{pmatrix}3\\-6\\3\end{pmatrix}=3\begin{pmatrix}1\\-2\\1\end{pmatrix}$ **and** $\begin{pmatrix}4\\-8\\4\end{pmatrix}=4\begin{pmatrix}1\\-2\\1\end{pmatrix}$ •2 ✓

$\therefore \overrightarrow{AB}=\frac{4}{3}\overrightarrow{BC}$

Ignore working subsequent to correct statement made on previous line.

$\Rightarrow$ AB is parallel to BC and B is a common point

$\Rightarrow$ A, B and C are collinear •3 ✓

Question	Generic scheme	Illustrative scheme	Max mark
(b)	$\bullet^4$ state ratio	$\bullet^4$ 3:4	1

Notes:

6. Answers in (b) must be consistent with the components of the vectors in (a) or the comparison of the vectors in (a). See Candidates C and D.
7. In this case, the answer for $\bullet^4$ must be stated explicitly in part (b).
8. The only acceptable variations for $\bullet^4$ must be related explicitly to AB and BC.
   For $\frac{BC}{AB}=\frac{4}{3}$, $\frac{AB}{BC}=\frac{3}{4}$ or $BC:AB=4:3$ stated in part (b) award $\bullet^4$. See Candidate E.
9. Accept unitary ratios for $\bullet^4$, eg $\frac{3}{4}:1$ or $1:\frac{4}{3}$.
10. Where a candidate states multiple ratios which are not equivalent, award 0/1.

Commonly Observed Responses:

Candidate C – using components of vectors

(a) $\overrightarrow{AB}=\begin{pmatrix}3\\-6\\3\end{pmatrix}$ $\bullet^1$ ✓

$\overrightarrow{BC}=\begin{pmatrix}4\\-8\\4\end{pmatrix}$

$\overrightarrow{BC}=\frac{3}{4}\overrightarrow{AB}$ $\bullet^2$ ✗

(b) 3:4 $\bullet^4$ ✓

Candidate D – using comparison of vectors

(a) $\overrightarrow{AB}=\begin{pmatrix}3\\-6\\3\end{pmatrix}$ $\bullet^1$ ✓

$\overrightarrow{BC}=\begin{pmatrix}4\\-8\\4\end{pmatrix}$

$\overrightarrow{BC}=\frac{3}{4}\overrightarrow{AB}$ $\bullet^2$ ✗

(b) 4:3 $\bullet^4$ ✓ 1

Candidate E – acceptable variation

$\frac{AB}{BC}=\frac{3}{4}$ $\bullet^4$ ✓

Ratio $=4:3$

Ignore working subsequent to correct statement made on previous line.

Candidate F – trivial ratio

Ratio is 1:1 $\bullet^4$ ✓ 2

Question			Generic scheme	Illustrative scheme	Max mark
6.			•1 write in differentiable form •2 start to differentiate •3 complete differentiation	•1 $(1-3x)^{-5}$ stated or implied by •2 •2 $-5(1-3x)^{-6}\ldots$ •3 $\ldots\times(-3)$	**3**

Notes:

1. Where candidates attempt to expand $(1-3x)^{-5}$, no further marks are available.
2. •2 is only available for differentiating an expression with a negative power.

Commonly Observed Responses:

Candidate A

$y=(1-3x)^{-5}$ •1 ✓

$\frac{dy}{dx}=-5(1-3x)^{-6}\times-3$ •2 ✓ •3 ✓

$\frac{dy}{dx}=-15(1-3x)^{-6}$

Candidate B

$y=(1-3x)^{-5}$ •1 ✓

$\frac{dy}{dx}=-15(1-3x)^{-6}$ •2 ✓ •3 ✗

Candidate C

$y=(1-3x)^{-5}$ •1 ✓

$\frac{dy}{dx}=-5(1-3x)^{-6}\times-3$ •2 ✓ •3 ✗

Candidate D - differentiating over two lines

$y=(1-3x)^{-5}$ •1 ✓

$\frac{dy}{dx}=-5(1-3x)^{-6}$ •2 ✓ •3 ^

$\frac{dy}{dx}=15(1-3x)^{-6}$

Question	Generic scheme	Illustrative scheme	Max mark
7.	**Method 1** •1 use $m = \tan\theta$ •2 find gradient of L •3 use property of perpendicular lines •4 determine equation of line	**Method 1** •1 $m = \tan 30°$ •2 $\frac{1}{\sqrt{3}}$ •3 $-\sqrt{3}$ •4 $y = -\sqrt{3}x - 4$	**4**
	Method 2 •1 find angle perpendicular line makes with the positive direction of the x-axis. •2 use $m = \tan\theta$ •3 find gradient of perpendicular line •4 determine equation of line	**Method 2** •1 $30° + 90° = 120°$ stated or implied by •2 •2 $m = \tan 120°$ •3 $-\sqrt{3}$ •4 $y = -\sqrt{3}x - 4$	

Notes:

1. In Method 1, where candidates make no reference to a trigonometric ratio or use an incorrect trigonometric ratio, •1 and •2 are unavailable.
   In Method 2, where candidates use an incorrect trigonometric ratio •2 and •3 are unavailable.
2. Accept $y + 4 = -\sqrt{3}(x)$ at •4, but do not accept $y + 4 = -\sqrt{3}(x - 0)$.
3. In Method 1, •4 is only available if the candidate has attempted to use a perpendicular gradient.

Commonly Observed Responses:

Candidate A

$m = \frac{1}{\sqrt{3}}$ (with or without diagram) •1 ^ •2 ✓ 2

$m_{\perp} = -\sqrt{3}$ •3 ✓ 1

Candidate B

$m = \tan\theta$ (with or without diagram) •1 ^

$m = \frac{1}{\sqrt{3}}$ •2 ✓ 1

Candidate C

$m = \tan\theta = 30$ •1 ✗

$m = \frac{1}{\sqrt{3}}$ •2 ✓ 1

Candidate D

$m = \tan^{-1} 30$ •1 ✗

$m = \frac{1}{\sqrt{3}}$ •2 ✓ 1

Candidate E

$\tan 30 = \frac{1}{\sqrt{3}}$ •1 ^

$m_{\perp} = -\sqrt{3}$ •2 ✓ 1 •3 ✓ 1

Question			Generic scheme	Illustrative scheme	Max mark
8.	(a)		•1 state integral	•1 $\int_{-1}^{2}\left(-x^2+x+2\right)dx$	**1**

Notes:

1. Evidence for •1 may be appear in part (b). However, where candidates make no attempt to answer part (a), •1 is not available.
2. •1 is not available to candidates who omit the limits or 'dx'.
3. •1 is awarded for a candidates final expression for the area. However, accept $\int_{-1}^{2}\left(\left(x^2+2x+3\right)-\left(2x^2+x+1\right)\right)dx$ or $\int_{-1}^{2}\left(x^2+2x+3\right)dx-\int_{-1}^{2}\left(2x^2+x+1\right)dx$ without further working.
4. For $\int_{-1}^{2}x^2+2x+3-2x^2+x+1\,dx$, see Candidates A and B.

Commonly Observed Responses:

Candidate A

(a) $\int_{-1}^{2}x^2+2x+3-2x^2+x+1\,dx$

$\int_{-1}^{2}\left(-x^2+x+2\right)dx$ •1 ✓

Treat missing brackets as bad form as subsequent working is correct.

Candidate B

(a) $\int_{-1}^{2}x^2+2x+3-2x^2+x+1\,dx$

(b) $\int_{-1}^{2}\left(-x^2+x+2\right)dx$ •1 ✓

•1 awarded in part (b)

Candidate C – error in simplification

(a) $\int_{-1}^{2}\left(x^2+2x+3\right)-\left(2x^2+x+1\right)dx$

$\int_{-1}^{2}x^2+x+2\,dx$ •1 ✗

Question	Generic scheme	Illustrative scheme	Max mark
(b)	$\bullet^2$ integrate expression from (a)	$\bullet^2$ $-\frac{1}{3}x^3+\frac{1}{2}x^2+2x$	3
	$\bullet^3$ substitute limits	$\bullet^3$ $\left(-\frac{1}{3}(2)^3+\frac{1}{2}(2)^2+2(2)\right)$ $-\left(-\frac{1}{3}(-1)^3+\frac{1}{2}(-1)^2+2(-1)\right)$	
	$\bullet^4$ evaluate area	$\bullet^4$ $\frac{9}{2}$	

Notes:

5. Where a candidate differentiates one or more terms at $\bullet^2$ then $\bullet^2$, $\bullet^3$ and $\bullet^4$ are unavailable.
6. Do not penalise the inclusion of '$+c$' or the continued appearance of the integral sign.
7. Candidates who substitute limits without integrating any term do not gain $\bullet^3$ or $\bullet^4$.
8. Where a candidate arrives at a negative value at $\bullet^4$ see Candidates D and E.

Commonly Observed Responses:

Candidate D

Eg $\int_{-1}^{2}\left(x^2-x-2\right)dx$

$\vdots$

$=-\frac{9}{2}=\frac{9}{2}$ $\bullet^4$ ✗

However…

$=-\frac{9}{2}$, hence area is $\frac{9}{2}$. $\bullet^4$ ✓

Candidate E

Eg $\int_{2}^{-1}\left(-x^2+x+2\right)dx$

$\vdots$

$=-\frac{9}{2}$ cannot be negative so $\frac{9}{2}$ units² $\bullet^4$ ✗

However…

$=-\frac{9}{2}$, hence area is $\frac{9}{2}$. $\bullet^4$ ✓

Candidate F – not using expression from (a)

(a) $\int_{-1}^{2}x^2+2x+3\,dx$ $\bullet^1$ ✗

(b) $\int_{-1}^{2}\left(x^2+2x+3\right)-\left(2x^2+x+1\right)dx$

$=\left[-\frac{1}{3}x^3+\frac{1}{2}x^2+2x\right]_{-1}^{2}$ $\bullet^2$ ✓ 2

$=\left(-\frac{1}{3}(2)^3+\frac{1}{2}(2)^2+2(2)\right)$

$-\left(-\frac{1}{3}(-1)^3+\frac{1}{2}(-1)^2+2(-1)\right)$ $\bullet^3$ ✓ 1

$=\frac{9}{2}$ $\bullet^4$ ✓ 1

Question			Generic scheme	Illustrative scheme	Max mark
9.	(a)	(i)	•1 form an expression	•1 $p(2p+16)+(-2)(-3)+(4)(6)$	1
		(ii)	•2 equate scalar product to 0 •3 factorise •4 state values of p	•2 $p(2p+16)+(-2)(-3)+(4)(6)=0$ •3 $2(p+5)(p+3)$ •4 -5 and -3	3

Notes:

1. Evidence for •1 may appear in part (a)(ii).
2. The appearance of '$\mathbf{u}\cdot\mathbf{v}=0$' alone is insufficient for •2.
3. For •2 to be awarded '$=0$' must appear at •2 or •3.
4. Do not penalise the absence of the common factor at •3.

Commonly Observed Responses:

Candidate A - incorrect expression at •2

(i) $p(2p+16)+(-2)(-3)+(4)(6)$ •1 ✓

$=2p^2+16p+30$

$=p^2+8p+15$

(ii) $p^2+8p+15=0$ •2 ✗

$(p+5)(p+3)=0$ •3 ✓ 1

$p=-5, p=-3$ •4 ✓ 1

Candidate B - incorrect expression at •2

(i) $p(2p+16)+(-2)(-3)+(4)(6)$ •1 ✓

$=2p^2+16p+30$

(ii) $p^2+8p+15=0$ •2 ✗

$(p+5)(p+3)=0$ •3 ✓ 1

$p=-5, p=-3$ •4 ✓ 1

Candidate C - incorrect expression at •2

$p(2p+16)+(-2)(-3)+(4)(6)$ •1 ✓

$2p^2+16p+24=0$ •2 ✗

$2(p+6)(p+2)$ •3 ✓ 1

$p=-6, p=-2$ •4 ✓ 1

Candidate D

(i) $\mathbf{u}.\mathbf{v}=\begin{pmatrix}2p^2+16p\\6\\24\end{pmatrix}$ •1 ✗

(ii) $p(2p+16)+6+24=0$ •2 ✓

$2p^2+16p+30=0$

$(p+5)(p+3)=0$ •3 ✓

$p=-5, p=-3$ •4 ✓

Question			Generic scheme	Illustrative scheme	Max mark
	(b)		•5 interpret relationship •6 determine value of p	•5 $3(p)=2(2p+16)$ or $3\mathbf{u}=2\mathbf{v}$ or equivalent •6 -32	2
Notes:					
Commonly Observed Responses:					
Candidate E For parallel vectors $\theta=0°$ Using $\mathbf{u}.\mathbf{v}=\lvert\mathbf{u}\rvert\lvert\mathbf{v}\rvert\cos\theta$ $p(2p+16)+(-2)(-3)+(4)(6)=\sqrt{p^2+(-2)^2+4^2}\sqrt{(2p+16)^2+(-3)^2+6^2}$ •5 ✓ $p^2+64p+1024=0$ $p=-32$ •6 ✓					

Question			Generic scheme	Illustrative scheme	Max mark
10.	(a)		•[1] identify value of a	•[1] 3	**1**
Notes:					
Commonly Observed Responses:					
	(b)		•[2] identify value of k	•[2] -2	**1**
Notes:					
Commonly Observed Responses:					

Question	Generic scheme	Illustrative scheme	Max mark
11.	•1 start to integrate	•1 $\sin\left(3x-\frac{\pi}{6}\right)\ldots$	**4**
	•2 complete integration	•2 $\ldots\times\frac{1}{3}$	
	•3 substitute limits	•3 $\left(\frac{1}{3}\sin\left(3\times\frac{\pi}{9}-\frac{\pi}{6}\right)\right)$ $-\left(\frac{1}{3}\sin\left(3\times0-\frac{\pi}{6}\right)\right)$	
	•4 evaluate integral	•4 $\frac{1}{3}$	

Notes:

1. Where candidates make no attempt to integrate or start to integrate individual terms within the bracket or use another invalid approach eg $\sin\left(3x-\frac{\pi}{6}\right)^2$ or $\int\cos(3x)-\cos\left(\frac{\pi}{6}\right)dx$, award 0/4.
2. Do not penalise the inclusion of '$+c$' or the continued appearance of the integral sign after •1.
3. Candidates who work in degrees from the start cannot gain •1. However, •2, •3 and •4 are still available.
4. •1 may be awarded for the appearance of $\sin\left(3x-\frac{\pi}{6}\right)$ in the first line of working, however see Candidates B and D.
5. •4 is only available where candidates have considered both limits within a trigonometric function.
6. Where candidates use a mixture of degrees and radians, •3 is not awarded. However, •4 is still available.

Commonly Observed Responses:

Candidate A - using addition formula

$\int_0^{\frac{\pi}{9}}\left(\cos 3x\cos\frac{\pi}{6}+\sin 3x\sin\frac{\pi}{6}\right)dx$

$=\frac{1}{3}\sin 3x\times\frac{\sqrt{3}}{2}\ldots$ •1 ✓

$\ldots-\frac{1}{3}\cos 3x\times\frac{1}{2}$ •2 ✓

Candidate B - integrated over two lines

$\int_0^{\frac{\pi}{9}}\left(\cos\left(3x-\frac{\pi}{6}\right)\right)dx$

$=\sin\left(3x-\frac{\pi}{6}\right)$ •1 ✓

$=\frac{1}{3}\sin\left(3x-\frac{\pi}{6}\right)$ •2 ✗

Candidate C - integrated in part

$3\sin\left(3x-\frac{\pi}{6}\right)$ •1 ✓ •2 ✗

$3\sin\left(3\times\frac{\pi}{9}-\frac{\pi}{6}\right)-3\sin\left(0-\frac{\pi}{6}\right)$ •3 ✓ 1

3 •4 ✓ 1

Candidate D - integrated in part

$-\frac{1}{3}\sin\left(3x-\frac{\pi}{6}\right)$ •1 ✗ •2 ✓

$-\frac{1}{3}\sin\left(3\times\frac{\pi}{9}-\frac{\pi}{6}\right)+\frac{1}{3}\sin\left(0-\frac{\pi}{6}\right)$ •3 ✓ 1

$-\frac{1}{3}$ •4 ✓ 1

Question			Generic scheme	Illustrative scheme	Max mark
12.	(a)		•1 interpret notation •2 state expression for $f\left(g\left(x\right)\right)$	•1 $f\left(5-x\right)$ or $\frac{1}{\sqrt{g\left(x\right)}}$ •2 $\frac{1}{\sqrt{5-x}}$	**2**
Notes:					
1. For $\frac{1}{\sqrt{5-x}}$ without working, award both •1 and •2.					
Commonly Observed Responses:					
Candidate A $5-\frac{1}{\sqrt{x}}$ •1 ✗ •2 ✓ 1					
	(b)		•3 state range	•3 $x \geq 5$	**1**
Notes:					
2. Answer at •3 must be consistent with expression at •2. 3. For candidates who interpret $g\left(f\left(x\right)\right)$ as $f\left(g\left(x\right)\right)$, do not award •3.					
Commonly Observed Responses:					
Candidate B $5-\frac{1}{\sqrt{x}}$ •1 ✗ •2 ✓ 1 $x \leq 0$ •3 ✗					

Question			Generic scheme	Illustrative scheme	Max mark
13.	(a)	(i)	$\bullet^1$ determine $\cos p$	$\bullet^1$ $\frac{2}{\sqrt{5}}$	**1**
		(ii)	$\bullet^2$ determine $\cos q$	$\bullet^2$ $\frac{3}{\sqrt{10}}$	**1**

Notes:

1. Where candidates do not simplify the perfect squares see Candidates A and B.

Commonly Observed Responses:

Candidate A – no evidence of simplification

$\cos p = \frac{\sqrt{4}}{\sqrt{5}}$ $\bullet^1$ ✖

$\cos q = \frac{\sqrt{9}}{\sqrt{10}}$ $\bullet^2$ ✓ 1

Repeated error not penalised twice

Candidate B – simplification in part (b)

(a) $\cos p = \frac{\sqrt{4}}{\sqrt{5}} \cos q = \frac{\sqrt{9}}{\sqrt{10}}$

$\bullet^1$ ✓ $\bullet^2$ ✓

$\vdots$

(b) $\sin(p+q) = \frac{5}{\ldots}$

Roots have been simplified in (b)

Question			Generic scheme	Illustrative scheme	Max mark
	(b)		$\bullet^3$ select appropriate formula and express in terms of p and q	$\bullet^3$ $\sin p \cos q + \cos p \sin q$	**3**
			$\bullet^4$ substitute into addition formula	$\bullet^4$ $\frac{1}{\sqrt{5}} \times \frac{3}{\sqrt{10}} + \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{10}}$	
			$\bullet^5$ evaluate $\sin(p+q)$	$\bullet^5$ $\frac{1}{\sqrt{2}}$	

Notes:

2. Award $\bullet^3$ for candidates who write $\sin\left(\frac{1}{\sqrt{5}}\right) \times \cos\left(\frac{3}{\sqrt{10}}\right) + \cos\left(\frac{2}{\sqrt{5}}\right) \times \sin\left(\frac{1}{\sqrt{10}}\right)$. $\bullet^4$ and $\bullet^5$ are unavailable.
3. For any attempt to use $\sin(p+q) = \sin p + \sin q$, $\bullet^4$ and $\bullet^5$ are unavailable.
4. At $\bullet^5$, accept answers such as $\frac{5}{\sqrt{50}}$ or $\frac{5}{5\sqrt{2}}$ but not $\frac{5}{\sqrt{5} \times \sqrt{10}}$.
5. At $\bullet^5$, the answer must be given as a single fraction.
6. Do not penalise trigonometric ratios which are less than -1 or greater than 1.

Commonly Observed Responses:

Question			Generic scheme	Illustrative scheme	Max mark
14.	(a)		•1 apply $m\log_n x = \log_n x^m$ •2 apply $\log_a x + \log_a y = \log_a xy$ •3 evaluate logarithm	•1 $\ldots\log_{10} 5^2$ stated or implied by •2 •2 $\log_{10}\left(4\times5^2\right)$ •3 2	**3**

Notes:

1. Each line of working must be equivalent to the line above within a valid strategy, however see Candidate A.
2. Do not penalise the omission of the base of the logarithm at •1 or •2.
3. Correct answer with no working, award 0/3.

Commonly Observed Responses:

Candidate A

$2\log_{10}(4\times5)$ •2 ✖

$2\log_{10}(20)$

$\log_{10}(20)^2$ •1 ✓ 1 •3 ^

Question	Generic scheme	Illustrative scheme	Max mark
(b)	**Method 1** •4 apply $\log_a x - \log_a y = \log_a \frac{x}{y}$ •5 express in exponential form •6 solve for x	**Method 1** •4 $\log_2 \frac{7x-2}{3} = \ldots$ •5 $\frac{7x-2}{3} = 2^5$ •6 14	3
	Method 2 •4 apply $m \log_n x = \log_n x^m$ •5 simplify •6 solve for x	**Method 2** •4 $\ldots = \log_2 2^5$ •5 eg $\log_2 \frac{7x-2}{3} = \ldots$ or $\log_2 (7x-2) = \log_2 (3 \times 2^5)$ •6 14	

Notes:

4. •6 is only awarded if each line of working is equivalent to the line above within a valid strategy.

Commonly Observed Responses:

Candidate A - invalid working leading to solution

$\log_2 \frac{7x-2}{3} = \log_2 5^2$ •4 ✓ •5 ✗

$x = 11$ •6 ✓ 2

Candidate B - invalid working leading to solution

$\log_2 \frac{7x-2}{3} = \log_2 5 \times 2$ •4 ✓ •5 ✗

$x = \frac{32}{7}$ •6 ✓ 2

Candidate C

$\log_2 \left(\frac{7x-2}{3} \right) = 5 \log_2 2$ •5 ✓

$\log_2 \frac{7x}{3} - \frac{2}{3} = \log_2 2^5$ •4 ✓

Candidate D

$\log_2 (7x-2) - \log_2 3 = \log_2 2^5$ •4 ✓

$\log_2 \left(\frac{7x-2}{3} \right) = \log_2 25$ •5 ✓

Question			Generic scheme	Illustrative scheme	Max mark
15.	(a)		•1 substitute appropriate double angle formula •2 factorise •3 solve for $\cos x°$ and $\sin x°$ •4 solve for x	•1 $2\sin x°\cos x°+6\cos x°=0$ •2 $2\cos x°(\sin x°+3)=0$ •3 $\cos x°=0$ •4 $\sin x°=-3$ •4 $x=90,\ 270$ 'no solutions'	**4**

Notes:

1. Do not penalise the absence of '$=0$' at •1 and •2.
2. Do not penalise the absence of '2' as a common factor at •2.
3. Do not penalise the omission of degree signs.
4. Candidates who leave their answer in radians do not gain •4 (if marking horizontally) or •3 (if marking vertically).
5. •4 is only available if one of the equations at •3 has no solution.
6. Accept ~~$\sin x° = -3$~~ at •4.

Commonly Observed Responses:

Candidate A

$2\sin x\cos x=-6\cos x$ •1 ✓

$2\sin x=-6$ •2 ^ •3 ^

~~$\sin x=-3$~~ •4 ✓ 1

Candidate B - insufficient evidence for •3

$2\sin x°\cos x°+6\cos x°=0$ •1 ✓

$2\cos x°(\sin x°+3)=0$ •2 ✓

$2\cos x°=0$, $\sin x°=-3$ •3 ^ •4 ^

However,

$x=90,\ 270$, 'no solutions' •3 ✓ •4 ✓

	(b)		•5 state solutions	•5 45, 135, 225,315	**1**

Notes:

Commonly Observed Responses:

Question			Generic scheme	Illustrative scheme	Max mark
16.	(a)		•1 identify centre •2 apply distance formula and demonstrate result	•1 (1, −2) stated or implied by •2 •2 $\sqrt{(4-1)^2+(k-(-2))^2}$ leading to $\sqrt{k^2+4k+13}$	**2**
Notes:					
1. Beware of candidates who 'fudge' their working between •1 and •2.					
Commonly Observed Responses:					
	(b)		•3 interpret information •4 express inequality in standard quadratic form •5 determine zeros of quadratic expression •6 state range with justification	•3 $\sqrt{k^2+4k+13}>5$ •4 $k^2+4k-12>0$ •5 −6, 2 •6 $k<-6, k>2$ with eg sketch or table of signs	**4**
Notes:					
2. Where a candidate has used an incorrect expression from part (a), •3 is not available. However, •4, •5 and •6 are still available for dealing with an expression of equivalent difficulty. 3. Candidates who do not work with an inequation from the outset lose •3, •4 and •6. However, •5 is still available. See Candidate A.					
Commonly Observed Responses:					
Candidate A $\sqrt{k^2+4k+13}=5$ •3 ✖ $k^2+4k-12=0$ •4 ✖ $k=-6,\ k=2$ •5 ✓ For P to lie outside the circle $k<-6,\ k>2$ •6 ✖					

Question			Generic scheme	Illustrative scheme	Max mark
17.	(a)		•1 expand brackets •2 use double angle formula for sin •3 use trigonometric identity and express in required form	•1 $\sin^2 x - \sin x \cos x$ $-\sin x \cos x + \cos^2 x$ •2 $\ldots - \sin 2x \ldots$ •3 $1 - \sin 2x$	**3**
Notes:					
1. For correct answer with no working award 0/3.					
Commonly Observed Responses:					
Candidate A – incorrect notation $\sin x^2 - 2\sin x \cos x + \cos x^2$ •1 ✗ $1 - \sin 2x$ •2 ✓ •3 ✗					
	(b)		•4 link to (a) and integrate one term •5 complete integration	•4 eg $\int (1 - \sin 2x)\, dx = x \ldots$ •5 $x + \frac{1}{2}\cos 2x + c$	**2**
Notes:					
2. •4 and •5 can only be awarded if the integrand is of the form $p + q \sin rx$. 3. Where the statement for •3 appears with no relevant working, •4 and •5 are not available.					
Commonly Observed Responses:					

[END OF MARKING INSTRUCTIONS]